



A guide for selecting periodic water wave theories - Le Méhauté (1976)'s graph revisited

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ABSTRACT

This note provides guidelines for selecting appropriate analytical periodic water wave solutions for applications, based on two physical parameters, namely, frequency dispersion parameter (water depth divided by wavelength) and the nonlinearity parameter (wave height divided by wavelength). The guidelines are summarized in a graphic format in the two-parameters space. The new graph can be viewed as the quantification of the well-known Le Méhauté (1976)'s graph with quantitative demarcations between applicable analytical periodic water wave solutions. In the deep water and intermediate water regimes the fifth-order Stokes wave theories (Zhao and Liu, 2022) are employed for the construction of the graph, while in the shallower water regime, Fenton (1999)'s higher order cnoidal wave theories are used. The dividing lines between the applicable ranges of Stokes wave and cnoidal wave theories are determined by the values of Ursell number, a ratio between the nonlinearity and frequency dispersion.

1. Introduction

In coastal/ocean engineering projects, based on given wave conditions, it is often desirable to first determine the appropriate wave theory to be used for analyses/design. Similar needs exist in conducting research for setting up laboratory experiments or numerical simulations. Within the framework of periodic waves, Le Méhauté (1976) first provided such useful guidelines in graphic format (it will be referred as the graph hereafter), showing the range of validity of Stokes and cnoidal wave theories in terms of two dimensional parameters: H/T^2 and h/T^2 , where T represents the wave period, H denotes the wave height, and h is the still water depth. The two-parameters space, covering from deep water to shallow water, is bounded by a wave breaking criterion. Dividing lines were drawn to specify the parameter spaces for different order of Stokes wave solutions and cnoidal wave solutions, derived from perturbation methods. The graph has been widely referred to in coastal engineering literature. Some of the recent published journal articles include, but not limited to He et al. (2023), Zhang and Benoit (2021), Samak et al. (2021), Li et al. (2021), Izquierdo et al. (2021), Eldrup and Andersen (2019), and Mohseni et al. (2018), covering a wide range

of applications. The graph has also been introduced in textbooks such as Holthuijsen (2010), and Kamphuis (2000), and coastal engineering protection manual and design guidelines (Coastal Engineering Research Center (US), 1973; US Army Corps of Engineers, 2002). Therefore, the graph has had a deep impact on coastal engineering education, research and application. However, Le Méhauté (1976) cautioned that “the graph is somewhat arbitrary and merely qualitative” because “a comprehensive quantitative investigation of error, which is made by using various theories in various domains, has not been done so far”. Therefore, there is need to quantify the graph with correct periodic wave theory solutions.

Over the last half centuries, several analytical forms of high-order (up to 5th order) Stokes wave solutions have been presented in the literature (De, 1955; Skjelbreia and Hendrickson, 1960; Cokelet, 1977b; Fenton, 1985). A detailed review of the differences among them has been provided in Zhao and Liu (2022), in which a set of analytical Stokes wave solutions up to fifth order has also been derived. Note that the Stokes expansions can be formulated in terms of potential function or stream function, which are related through the velocity definitions. The alternatives to the analytical solutions, based on the

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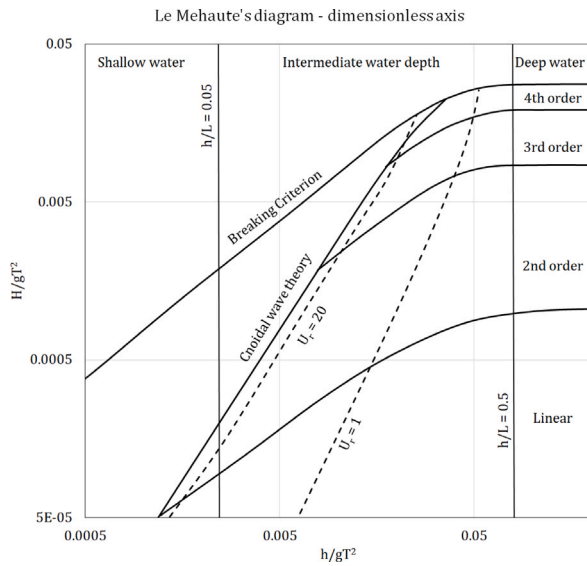


Fig. 1. Reproduction of Le Méhauté (1976)'s graph. The original graph has been re-scaled and re-plotted in terms of dimensionless parameters, $H/(gT^2)$ and $h/(gT^2)$.

Stokes expansions, are the numerical solutions employing the Fourier series expansion (e.g., Dean (1965), Rienecker and Fenton (1981), Sobey et al. (1987), Fenton (1988) and Ducroz et al. (2019)). Most of them are formulated in terms of stream functions and are often called stream function solutions (theories). Moreover, since they are not perturbation solutions, the order of solutions is actually referred to the highest harmonics included in the solutions.

To further discuss the graph's major features, it is reproduced in Fig. 1, using the dimensionless coordinates suggested in the Coastal Engineering Manual (CEM) (US Army Corps of Engineers, 2002). The coordinates of the graph were re-scaled by the gravitational acceleration, g , and the original graph was re-sketches in terms of two dimensionless parameters: $H/(gT^2)$ and $h/(gT^2)$ (see Figure II-1-20 in US Army Corps of Engineers (2002)). Although the range of validity of the Stokes wave theories was discussed quantitatively in the CEM (see Equations (II-1-112) and (II-1-113) in US Army Corps of Engineers (2002)), these criteria were not implemented in the graph. On the other hand, the parameter space for the Fourier (numerical) solutions, based on the stream function formulation (Dean, 1974), was added in the CEM's graph, which was not included in Le Méhauté's original graph.

The upper bound of the parameter space in Fig. 1 is determined by wave breaking criteria. In deep water condition, the breaking wave height is only related to the wavelength, L , while in shallow water, it is limited by the water depth (Galvin, 1972). The curve for breaking criteria is plotted using the formula by Miche (1944), which covers the entire range of water depth from deep to shallow (Galvin, 1972). In the deep water limit, the wave breaking curve approaches $(H/L)_B = 0.14$ (Miche, 1944; Galvin, 1972; Le Méhauté, 1976; Fenton, 1990).

The Ursell number is defined as $U_r = (H/L)/(h/L)^3$, measuring the relative importance of nonlinearity and frequency dispersion. In Fig. 1, a line representing $U_r = 26$ is plotted to divide the parameter spaces between cnoidal wave theories and Stokes wave theories. As suggested in Le Méhauté (1976), Stokes wave theories should be employed in the parameter space to the right of $U_r = 26$ line, where sub-parameter-spaces for using different orders of Stokes wave theory are defined based on $H/(gT^2)$ values in the deep water limit. However, the shapes of these dividing lines are arbitrary as stated in Le Méhauté (1976). On the other side of $U_r = 26$ line, the cnoidal wave theory was recommended in Le Méhauté (1976) without further sub-dividing the parameter space for different orders of cnoidal wave theories. Note that

the lines denoting $U_r = 1$ and $U_r = 20$, appearing in the graph, are kept in Fig. 1.

The objective of this note is to quantify (Le Méhauté, 1976)'s original graph for Stokes wave and cnoidal wave theories in the two dimensionless physical parameter space, i.e., H/L and h/L . In particular, Zhao and Liu (2022)'s fifth-order Stokes wave solutions, derived in terms of potential function, are employed in the analyses. Zhao and Liu (2022)'s derivation employed the Stokes expansions, resulting in closed form solutions. Similarly, employing the Stokes expansions (Fenton, 1985), formulated the Stokes wave problem in terms of stream function. In obtaining the high order solutions, Fenton used a non-physical assumption, requiring the wave height, H , to be twice of the amplitude of the first-order first harmonic wave solution, a , i.e. $H = 2a$. In fact, as shown in Zhao and Liu (2022) the relationship between H and a is a part of the solutions, depending on the order of the Stokes wave theories. To authors' best knowledge, all the existing Stokes expansion solutions in terms of stream function, including Kishida and Sobey (1988) and Sobey et al. (1987), adopted the same non-physical assumption, $H = 2a$, and, thus, suffered the same shortcomings as Fenton (1985)'s solutions in high orders. In this note, keeping with same spirit as the graph, we shall only focus on the closed form solutions based on the Stokes expansion and will not discuss the stream function solutions further.

The graph is redesigned in the following manner. Two dimensionless parameters, H/L and h/L , are used to define the parameter space for different periodic water wave theories. The definitions for each sub-parameter-space suitable for different wave theory are quantified. While the applicable ranges of different orders of Stokes wave solutions are based on Zhao and Liu (2022), the applicable ranges of different order of cnoidal wave theories are determined by the parameter of elliptic integral, m (Fenton, 1999).

The new graph is displayed in Fig. 2. It is noted here that the constant value line for $m = 0.96$ coincides closely with the constant value line for $U_r = 50$ and the $m = 0.855$ coincides closely with that for $U_r = 26$. In the remainder of this paper, the procedure for developing this new graph — Fig. 2 — is discussed. In Section 2.1, the free surface solutions and nonlinear dispersion relationship of Stokes waves to be used for further analysed are presented. The criteria for defining the sub-parameter spaces for different orders of Stokes wave theories are then quantified. The various orders of cnoidal wave theories are discussed in Section 2.2. The parameter space of periodical wave theories is capped with the curve representing breaking wave limit, which is further discussed in Section 2.3. In Section 3 concluding remarks are provided. The Matlab figure and relevant codes on how to use the new figure for research and engineering applications are provided as follows. <https://github.com/KuifengZhao/waveModelSelection.git>.

2. Development of the new graph for selecting water wave theories

The procedure for developing Fig. 2 is explained herein. Two dimensionless parameters, H/L and h/L , are employed to define the parameter space suitable for different wave theories. As shown in Fig. 2, it is commonly accepted that the deep water regime is defined as $h/L > 1/2$; while the shallow water regime is defined as $h/L < 1/20$. The intermediate depth regime is defined as $1/20 < h/L < 1/2$. Generally speaking, the Stokes wave theories are not applicable in very shallow water regime unless the wave height is also very small in comparison with water depth (Hedges, 1995). On the other hand, the cnoidal wave theories become invalid in the deep water regime, or when the U_r number becomes too small (Fenton, 1990). Various authors have recommended different criteria for the dividing line between Stokes wave theories and cnoidal wave theories. For example, Holthuijsen (2010) and Dean and Dalrymple (1991) suggested that Stokes wave theories are applicable when $U_r \leq 10$. Le Méhauté (1976)'s original chart showed three values of U_r , and one is $U_r = 20$ without clearly

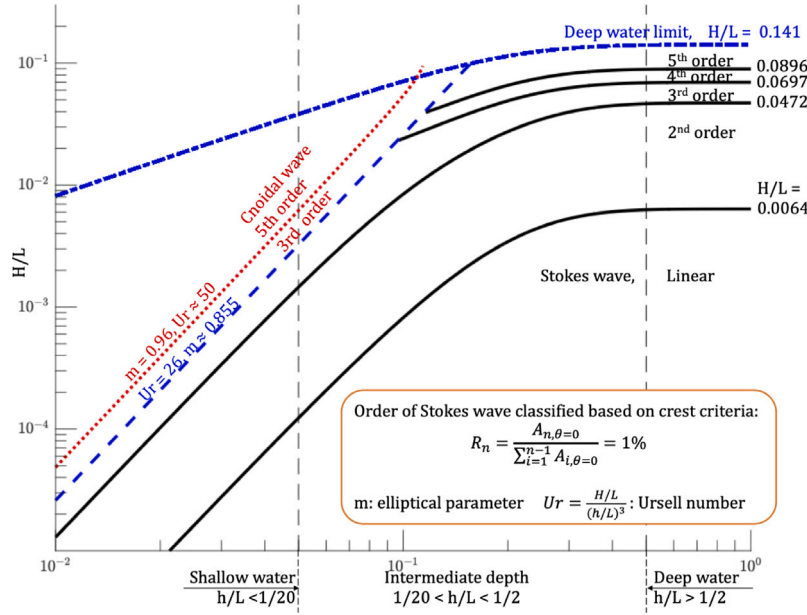


Fig. 2. New graph specifying the applicable range of different wave theories in the parameter space of h/L and H/L . The parameters in the graph are defined as follow: h is the water depth; H is the wave height; L is the wave length; m is the elliptical parameter; $U_r = (H/L)/(h/L)^3$ is the Ursell number; R_n is the crest criteria of Stokes wave.

indicating if it is boundary of Stokes wave and cnoidal wave. Coastal Engineering Research Center (US) (1973) adopted $U_r \geq 26$ as the parameter space for using cnoidal wave theories; Holthuijsen (2010) also suggested that both Stokes wave and cnoidal wave theories apply equally well when $10 < U_r < 26$; Hedges (1995) and Fenton (1990) suggested an $U_r = 40$ line as the boundary.

In the new graph (Fig. 2), the $U_r = 26$ line is adopted as the boundary of the parameter space for Stokes wave theories. In other words, when $U_r > 26$, Stokes wave theories are not recommended; instead, cnoidal wave theories should be employed. On the other hand, the cnoidal theories are not recommended when $U_r \leq 26$. More detailed discussions are provided in following sections.

2.1. Parameter space for Stokes wave theories ($U_r \leq 26$)

In this section, we shall discuss how to define the demarcation lines between the parameter spaces of different orders of Stokes wave theories. In developing Fig. 2, Zhao and Liu (2022)'s closed-form solutions are used to quantify the sub-parameter spaces for different orders of Stokes wave theory. Following Zhao and Liu (2022), the free surface elevation of a Stokes wave, up to the fifth order, propagating in the x -direction with wave number k and frequency ω , can be expressed as

$$\eta/a = \sum_{i=1}^5 A_i + O[(\pi H/L)^6] \quad (1)$$

with

$$\begin{aligned} A_1 &= \cos \theta, \\ A_2 &= kaB_{22} \cos 2\theta, \\ A_3 &= k^2 a^2 (B_{31} \cos \theta + B_{33} \cos 3\theta), \\ A_4 &= k^3 a^3 (B_{42} \cos 2\theta + B_{44} \cos 4\theta), \\ A_5 &= k^4 a^4 (B_{51} \cos \theta + B_{53} \cos 3\theta + B_{55} \cos 5\theta), \end{aligned} \quad (2)$$

where a is the wave amplitude of the first order wave, and $\theta = kx - \omega t + \psi$ is the phase function with the initial phase ψ . The full set of Stokes wave solutions can be found in Zhao and Liu (2022). The coefficients, B_{**} , in the equation above, which depend on h/L , can be found in supplementary material.

According to Zhao and Liu (2022), the wave height $H (= \eta_{\theta=0} - \eta_{\theta=\pi})$, is related to wave amplitude and relative water depth as follows:

$$\frac{H}{2a} = 1 + (B_{31} + B_{33})k^2 a^2 + (B_{51} + B_{53} + B_{55})k^4 a^4. \quad (3)$$

Moreover, the nonlinear dispersion relationship is obtained as

$$\begin{aligned} \omega &= \omega_0 (1 + k^2 a^2 \omega_2 + k^4 a^4 \omega_4), \\ \omega_2 &= \frac{2\beta^2 + 7}{4(\beta - 1)^2}, \\ \omega_4 &= \frac{20\beta^6 + 132\beta^5 + 20\beta^4 - 184\beta^3 - 243\beta^2 - 11\beta + 428}{32(\beta - 1)^5(\beta + 1)}, \end{aligned} \quad (4)$$

where $\beta = \cosh 2kh$. For a given wave height, H , water depth, h , and wave period, $T = 2\pi/\omega$, Eqs. (3) and (4) can be solved simultaneously to find wave amplitude a and wave number k .

Calculating $A_{n,\theta=0}$, Eq. (2), at $\theta = 0$ yields the contribution of the n th order Stokes wave solution to the wave crest, $A_{n,\theta=0} \cdot \sum_{i=1}^{n-1} A_{i,\theta=0}$ denotes the summation of the contributions of the preceding $(n - 1)$ terms on the crest, and the ratio R_n ,

$$R_n = \frac{A_{n,\theta=0}}{\sum_{i=1}^{n-1} A_{i,\theta=0}}, \quad n = 2, \dots, 5 \quad (5)$$

measures the relative importance of the n th order Stokes wave solution.

In Fig. 2 the demarcation curves between two adjacent Stokes wave theories with different orders are defined by

$R_n = 1\%$, $n = 2, \dots, 5$. Note that from Eq. (2), A_n can be determined, once h/L and H/L values are specified. Thus, applying the root-finding method on Eq. (5), the four curves shown in Fig. 2 can be traced by specifying $R_n = 1\%$, $n = 2, \dots, 5$. It needs to be pointed out that the Stokes wave solutions are lower order compatible, i.e., one can always use Stokes 5th order solutions to calculate the wave properties in the range of a lower order Stokes wave, but not the other way around.

When the relative water depth, h/L , is much greater than 0.5 so that $\tanh kh \approx 1$, the formulas for linear dispersion relationship can be simplified and the wavelength is independent of water depth. This is the region where various orders of Stokes wave solutions depend only on nonlinearity parameter H/L . In other words, the dividing lines between Stokes wave theories become nearly horizontal. Their asymptotic values are listed in Table 1. In the intermediate water

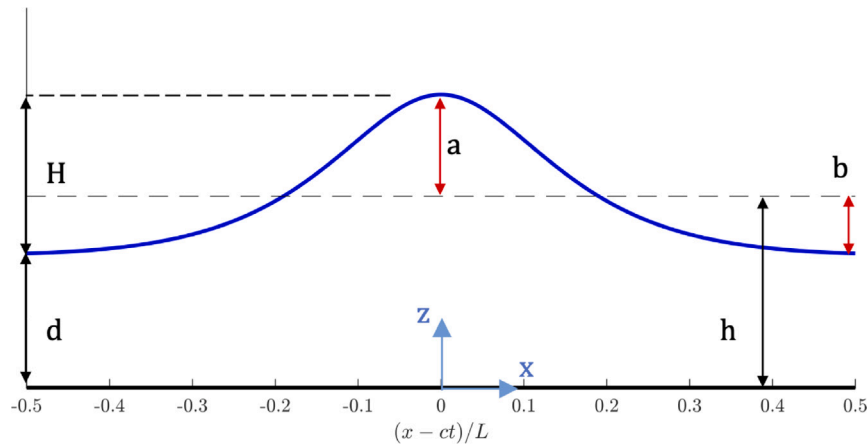


Fig. 3. A typical free surface profile of cnoidal wave.

Table 1

Deep water limit of H/L for various orders of Stokes wave.

Order of Stokes wave	Criteria	Deep water limit
Linear	$R_2 \leq 1\%$	$H/L \leq 0.0064$
Stokes II or higher	$R_2 \geq 1\%$	$H/L \leq 0.0472$
Stokes III or higher	$R_3 \geq 1\%$	$H/L \leq 0.0697$
Stokes IV or higher	$R_4 \geq 1\%$	$H/L \leq 0.0896$
Stokes V	$R_5 \geq 1\%$	$H/L \leq 0.141$

and the shallow water regimes, Stokes wave theories break down unless the nonlinearity parameter remains extremely small. As suggested in Coastal Engineering Research Center (US) (1973), the line, $U_r = 26$, marks the left side boundary of Stokes wave theories in the new graph (Fig. 2).

It is noted that the relative importance of the n th order Stokes wave solutions can be also evaluated by using the Fourier series solution (Ducroz et al., 2019) as the reference crest height, i.e., the denominator in Eq. (5). It is found that the resulting demarcation lines remain almost identical. In addition, interested readers could refer to Section 6 of the supplementary materials for a comparison of wave theories using the original and updated charts.

2.2. Parameter space for cnoidal wave theory ($U_r > 26$)

To facilitate further discussions on the parameter spaces where the cnoidal wave theories should be used, the solutions for free surface presented by Fenton (1999) for the 3rd order and 5th order cnoidal wave theories are briefly shown here:

$$\frac{\eta}{d} = 1 + \sum_{i=1}^j C_i, \quad (6)$$

where $j = 3$ or 5 for the 3rd and 5th order cnoidal wave theories, respectively. For brevity, the detailed expressions for C_i are given in supplementary material. A typical free surface profile of a cnoidal wave is shown in Fig. 3, in which $H = a + b$, being the distance from the trough to the crest, denotes the wave height. Note that the still water depth is h and the water depth below the trough is $d = h - b$. Basically, the free surface solutions are expressed as power series of Jacobi elliptic cosine function, $\text{cn}(\alpha(x - ct)/d|m)$, where $\alpha = H/d$, c is phase speed,

and m is the elliptical parameter. The relationships between wavelength L and m value are also available for the 3rd order and 5th order cnoidal wave theory in the supplementary material. These relationships are used to find the m value once the wave parameters, such as H, h, L , are provided.

As stated before, Coastal Engineering Research Center (US) (1973) suggested that the curve for $U_r = 26$ can be used to divide the parameter space between cnoidal wave theory and Stokes wave theory. To verify this suggestion, the Stokes wave theory solutions and the cnoidal theory solutions are computed and compared for $U_r = 26$. While the detailed comparisons can be found in supplementary material, the conclusions are summarized here. The differences in wave crest elevation of the free surface profiles between the 5th order Stokes wave theory and the 3rd order cnoidal wave theory is less than 0.2%. However, the stream-wise velocity components at the wave crest differs more than 11% between these two theories if the $H/h > 0.26$ or $h/L > 0.1$. Knowing that the Stokes wave theories degenerate as $h/L \rightarrow 0$, the cnoidal wave theory is preferred on the left side of $U_r = 26$.

As the U_r number decreases and $m \rightarrow 0$, it is well-known that the cn function reduces to the sinusoidal function (Abramowitz et al., 1988). Therefore, on the right side of $U_r = 1$, the first order cnoidal theory is also applicable, while the second order cnoidal theory is applicable in the parameter space between $U_r = 1$ and $U_r = 10$. Nevertheless, it is recommended here that the Stokes wave theory is employed on the right side of $U_r = 26$ since the calculations for the Stokes wave theory solutions are much more straightforward than those for the cnoidal wave theory solutions.

In summary, as shown in Fig. 2, the 5th order cnoidal wave theory should be used in the parameter space above the curve represented by $m = 0.96$ or $U_r > 50$; the 3rd order cnoidal wave theory should be used in the parameter space between the curve represented by $U_r = 26$ and the $m = 0.96$ curve or $U_r \approx 50$.

2.3. Wave breaking

When wave amplitude exceeds the breaking limit, the wave breaks and generates turbulence. Once the breaking occurs, the Stokes and cnoidal wave theories are no longer valid. The most frequently cited wave breaking limit is the Miche formula (Miche, 1944), which has the limits of $(H/h)_B = 0.78$ in shallow water and $(H/L)_B = 0.14$ in deep water. Fenton (1990) introduced a new breaking criterion as follows:

$$\frac{H}{L} = \frac{h}{L} \left(\frac{0.141063(h/L)^2 + 0.0095721(h/L) + 0.0077829}{(h/L)^3 + 0.0788340(h/L)^2 + 0.0317567(h/L) + 0.0093407} \right), \quad (7)$$

which covers the all range of h/L . In the shallow water limit, it shows the breaking condition of $(H/h)_B = 5/6$ (Cokelet, 1977a) and in the deep water region, the breaking criteria becomes $(H/L)_B = 0.14$ in the deep region, which is the same as (Miche, 1944)'s formula. For more about wave breaking criteria, interested readers can refer to Munk (1949), Goda (1970), Hunter and Vanden-Broeck (1983), and Kamphuis (1991).

3. Conclusion remarks

In this paper, a guideline for selecting appropriate periodic water wave theories for applications, in a graphic format (see Fig. 2), in terms of two dimensionless parameters: H/L and h/L . The Stokes wave solutions derived by Zhao and Liu (2022) and cnoidal wave solutions by Fenton (1999) are used in the analyses. Each sub-parameter-space suitable for different periodic wave theory have been quantified. The dividing line between the applicable ranges of Stokes wave and cnoidal wave theories are determined by the value of Ursell number. In the guideline, the boundary is represented by $U_r = 26$.

The MatLab codes for calculating wavelength (when the wave period T is given) and for producing Fig. 2 can be downloaded from <https://github.com/KuifengZhao/waveModelSelection.git>.

Alternative criteria may be used to define the demarcations lines depending on the context of applications of the wave theory. For example, readers may be more interested in velocities or accelerations when investigating the Morrison loads. More careful thoughts and analyses are required to assess if velocities or accelerations are suitable to obtain demarcation lines. Further study could be conducted in the future.

CRedit authorship contribution statement

Kuifeng Zhao: Methodology, Coding for Stokes wave, Preliminary draft and response to comments. **Yufei Wang:** Cnoidal wave solutions, Update manuscript and response to comments. **Philip L.-F. Liu:** Conceptualization, Structure of manuscript, Update manuscript and response to comments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.coastaleng.2023.104432>.

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